Lecture 7 v2: Functional Forms

25117 - Econometrics

Universitat Pompeu Fabra

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What we learned in the last lesson

- Hypothesis tests and confidence intervals for a single regression coefficient are carried out using essentially the same procedures used in the simple linear regression model. For example, a 95% confidence interval for $\hat{\beta}_1$ is given by $\hat{\beta}_1 \pm 1.96SE(\hat{\beta}_1)$

- Hypotheses involving more than one restriction on the coefficients are called joint hypotheses. Joint hypotheses can be tested using an F-statistic. F-statistics relate to $F_{q,n-k-1}$ distributions.

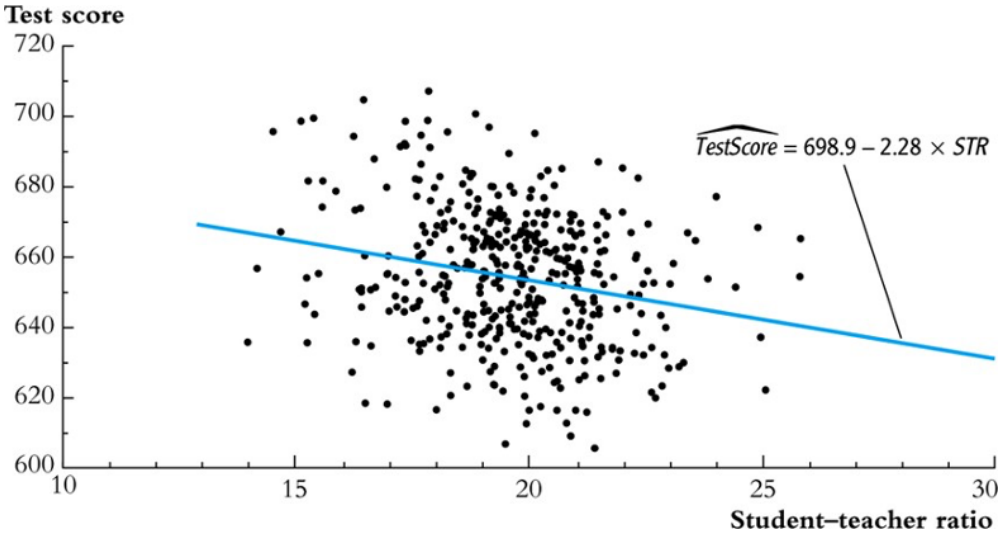
- Regression specification proceeds by first determining a base specification chosen to address concern about omitted variable bias. The base specification can be modified by including additional regressors that control for other potential sources of omitted variable bias.

- Simply choosing the specification with the highest R^2 can lead to regression models that do not estimate the causal effect of interest.

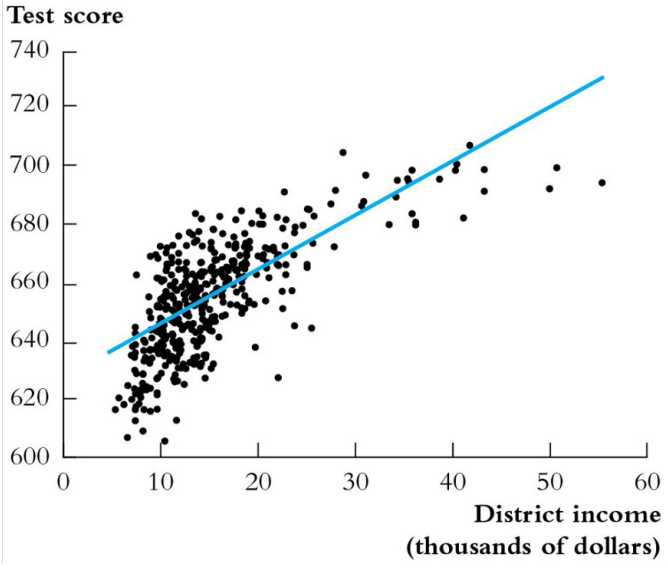
Introduction

- The regression function so far has been linear in the X's... But the linear approximation is not always a good one!
- The multiple regression model can handle regression functions that are nonlinear in one or more X.
- If a relation between Y and X is nonlinear:
  - The effect on Y of a change in X depends on the value of X – that is, the marginal effect of X is not constant
  - A linear regression is misspecified: the functional form is wrong
  - The estimator of the effect on Y of X is biased: in general, it isn't even right on average.
  - The solution is to estimate a regression function that is nonlinear in X

A linear fit on a (possibly) linear relationship



A linear fit on a non-linear relationship



The general nonlinear population regression function

The general nonlinear population regression function is defined as

$$Y_i = f(X_{1i}, X_{2i}, \ldots, X_{ki}) + u_i$$

Assumptions:

- $\mathbb{E}(u_i \mid X_{1i}, X_{2i}, \ldots, X_{ki}) = 0$ so $f(.)$ is the conditional expectation of Y given the X's.
- $(Y_i, X_{1i}, X_{2i}, \ldots, X_{ki})$ are i.i.d.
- Big outliers are rare (same idea as with linear OLS; the precise mathematical condition depends on the specific $f(.)$).
- No perfect multicollinearity (same idea; the precise statement depends on the specific $f(.)$).

The expected difference in Y associated with a difference in X_1, ceteris paribus, is

$$\Delta Y = f(X_1 + \Delta X_1, X_2, \ldots, X_k) - f(X_1, X_2, \ldots, X_k)$$

Nonlinear Functions of a Single Independent Variable

We'll look at two complementary approaches:

- **Polynomials in X**
  - The population regression function is approximated by a quadratic, cubic, or higher-degree polynomial

- **Logarithmic transformations**
  - Y and/or X is transformed by taking its logarithm, which provides a "percentages" interpretation (i.e., elasticities and semi-elasticities) of the coefficients that makes sense in many applications

The choice of specification (functional form) should be guided by judgment (which interpretation makes the most sense in your application?), tests, and plotting predicted values

Polynomials in X

Approximate the population regression function by a polynomial:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \beta_3 X_i^3 + \cdots + \beta_r X_i^r + u_i$$

- This is just the linear multiple regression model – except that the regressors are powers of X!
- Estimation, hypothesis testing, etc. proceeds as in the multiple regression model using OLS
- The coefficients are difficult to interpret, but the regression function itself is interpretable

For example, let's estimate the following quadratic specification:

$$\text{TestScore}_i = \beta_0 + \beta_1 \text{Income}_i + \beta_2 \text{Income}_i^2 + u_i$$

Polynomials in X

Dependent variable: TestScore

Regressor	Coefficient
Income	3.850995
Income^2	-0.0423085
Intercept	607.3017

$n = 420$ $R^2 = 0.5562$ SER $= 12.724$

Income is district per capita income, measured in thousands of dollars.

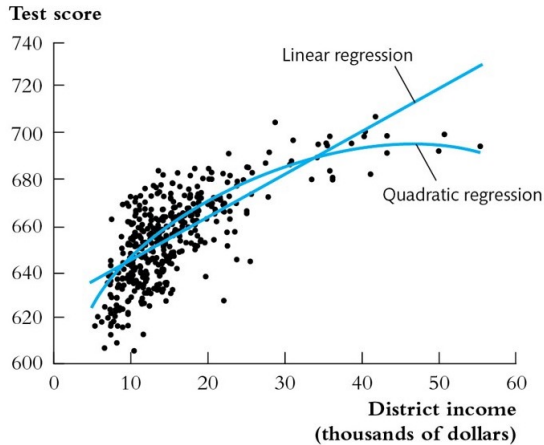
Polynomials in X

- We can plot the predicted values...
- The predicted change in *TestScore* for a change in income from \$5K per capita to \$6K per capita is :

  $(6 \times 3.85 - 6^2 \times 0.042) - (5 \times 3.85 - 5^2 \times 0.042) = 3.4$

$\Delta Income$	$\Delta TestScore$
from 5 to 6	3.4
from 25 to 26	1.7
from 45 to 46	0.0

- Caution! What is the effect of a change from \$100K to \$101K?



- The "effect" of a change in income is greater at low than high income levels (i.e., a declining marginal effect of income)

Polynomials in X

Dependent variable: TestScore

Regressor	Coefficient
Income	5.018677
Income^2	−0.0958052
Income^3	0.0006855
Intercept	600.079

$n = 420$ $R^2 = 0.5584$ $\text{SER} = 12.707$

Income is district per capita income, measured in thousands of dollars.

Polynomials in X

Link to Lecture 6: restricted versus unrestricted regression.

$$H_0 : \beta_2 = \beta_3 = 0 \qquad \text{vs.} \qquad H_1 : \beta_2 \neq 0 \text{ and/or } \beta_3 \neq 0.$$

Unrestricted (U): estimate all four coefficients in the cubic model:

$$\text{TestScore}_i = \beta_0 + \beta_1 \text{Income}_i + \beta_2 \text{Income}_i^2 + \beta_3 \text{Income}_i^3 + u_i.$$

Restricted (R): impose H_0 and re-estimate the linear model:

$$\text{TestScore}_i = \beta_0 + \beta_1 \text{Income}_i + u_i.$$

Thus $SSR_R = SSR_{\text{linear}}$ and $SSR_U = SSR_{\text{cubic}}$, using the same observations.

$$\underbrace{F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k_U - 1)}}_{\text{Lecture 6}} = \frac{(SSR_{\text{linear}} - SSR_{\text{cubic}})/2}{SSR_{\text{cubic}}/(n-4)}.$$

$q = 2$ restrictions; $k_U = 3$ regressors (excluding the intercept), so $n - k_U - 1 = n - 4$.

Reject H_0 at 1% if F exceeds the $F_{2,n-4}$ critical value, or if $p < 0.01$.

Log Changes: Why the Approximation Works

Throughout these slides, log denotes the natural logarithm; $Y > 0$.

1. Approximate $\log(x)$ near $x = 1$.

$$f(x) \approx f(1) + f'(1)(x-1).$$

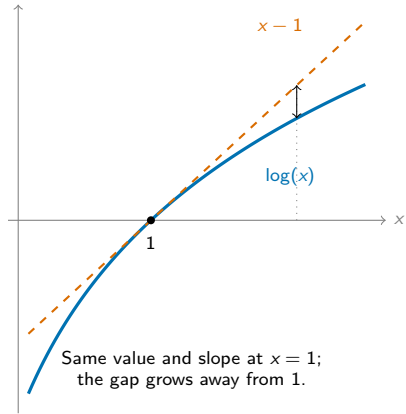
For $f(x) = \log(x)$, $f(1) = 0$ and $f'(1) = 1$, so

$$\boxed{\log(x) \approx x - 1 \quad \text{near } x = 1.}$$

2. Let x be the ratio of new to old Y:

$$x = \frac{Y + \Delta Y}{Y} = 1 + \frac{\Delta Y}{Y}.$$

A small proportional change makes x close to 1.



3. Use the log-ratio identity and multiply by 100:

$$\log(Y + \Delta Y) - \log(Y) = \log\left(1 + \frac{\Delta Y}{Y}\right) \approx \frac{\Delta Y}{Y} \quad \Rightarrow \quad 100\,\Delta \log Y \approx \text{percentage change in } Y.$$

Log Changes versus Percentage Changes

Define two different measures of the same change:

$$g = \frac{Y_{\text{new}} - Y_{\text{old}}}{Y_{\text{old}}}, \qquad d = \log(Y_{\text{new}}) - \log(Y_{\text{old}}) = \log(1+g).$$

$100g$ is the percentage change; $100d$ is the change in **log points**.

$$\boxed{g = e^d - 1 \quad \text{(exact)}} \qquad \boxed{g \approx d \quad \text{(only for small changes)}}.$$

Change in Y	Actual change: $100g$	Log points: $100\log(1+g)$
$100 \to 101$	1.00%	1.00
$100 \to 105$	5.00%	4.88
$0.7 \to 1.2$	71.43%	53.90

Our example is a large change:

$$g = \frac{1.2 - 0.7}{0.7} = 0.7143, \qquad d = \log(1.2/0.7) = 0.5390 \approx 0.54.$$

Thus about 54 log points correspond to a **71.4% increase**, not a 54% increase.
The approximation understates the increase by about 17.5 percentage points here.

Why About 54 Successive Increases of 1%?

1. Each 1% increase multiplies the current value by 1.01**.** After m such increases, $Y_{\text{new}} = Y_{\text{old}}(1.01)^m$.

$$\log(Y_{\text{new}}/Y_{\text{old}}) = m\log(1.01) \quad \Rightarrow \quad m = \frac{\log(1.2/0.7)}{\log(1.01)} = \frac{0.5390}{0.009950} \approx 54.17.$$

2. This explains the number 54: each 1% step adds about 0.01 to the log.

$$0.7(1.01)^{54} \approx 1.198 \approx 1.2, \qquad 100[(1.01)^{54} - 1] \approx 71.14\%.$$

The increases compound: later 1% increases apply to a larger base. Simply adding $54 \times 1\% = 54\%$ misses this growth of the base.

3. More generally, logarithms turn compounded growth into a sum. If $R_t = (Y_t - Y_{t-1})/Y_{t-1}$ is the proportional growth rate in period t,

$$\frac{Y_T}{Y_0} = \prod_{t=1}^{T}(1 + R_t) \quad \Rightarrow \quad \log(Y_T) - \log(Y_0) = \sum_{t=1}^{T}\log(1 + R_t).$$

This equality is exact. For small period-by-period changes, $\log(1 + R_t) \approx R_t$.

Logarithmic functions of Y and/or X

In the linear regression model,

$$Y_i = \beta_0 + \beta_1 X_{1i} + u_i$$

1-unit increase in X_{1i} was associated with a β_1-unit increase in Y_i.

Now, depending on the hypothesis we derived from the theory, data can be log-transformed on the RHS or the LHS or both.

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 log(X_{1i}) + u_i \\ log(Y_i) &= \beta_0 + \beta_1 X_{1i} + u_i \\ log(Y_i) &= \beta_0 + \beta_1 log(X_{1i}) + u_i \end{aligned}$$

In each case, β_1 has a different interpretation.

Linear-log: Rising but Flattening

Model and change (holding u fixed):

$$Y = \beta_0 + \beta_1 \log X + u,$$

$$\Delta Y = \beta_1[\log(X + \Delta X) - \log X] \approx \beta_1 \frac{\Delta X}{X}.$$

A 1% increase in X:

$$\frac{\Delta X}{X} = 0.01 \quad \Rightarrow \quad \Delta Y \approx \frac{\beta_1}{100}.$$

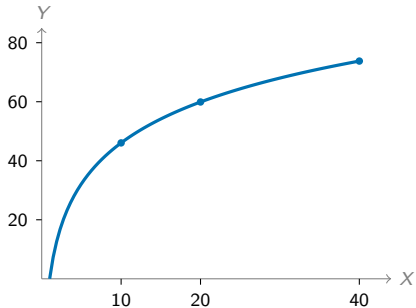
This is a change in **units of** Y, not percent. With $\beta_1 = 20$, it is about 0.2 units.

Why does the curve flatten?

$$\boxed{\frac{dY}{dX} = \frac{\beta_1}{X}.}$$

For $\beta_1 > 0$, a larger X means a smaller slope: the same extra unit of X adds less to Y.

Example: $Y = 20 \log X$



X	10	20	40
Slope $20/X$	2	1	0.5

Doubling X always adds $20 \log 2 \approx 13.86$ units to Y, but requires a larger dollar increase at higher X.

Graphs use the original X, Y axes and set $u = 0$. With $\beta_1 < 0$, this curve decreases and becomes less steep.

Log-linear: Constant Percentage Growth

Model and change (holding u fixed):

$$\log Y = \beta_0 + \beta_1 X + u, \qquad \Delta \log Y = \beta_1 \Delta X.$$

For a small log change, $\Delta \log Y \approx \Delta Y / Y$.

A one-unit increase in X:

$$\Delta X = 1 \;\Rightarrow\; \frac{\Delta Y}{Y} \approx \beta_1 \;\Rightarrow\; 100\frac{\Delta Y}{Y} \approx 100\beta_1.$$

Multiplying by 100 gives the **percentage change in Y**.

With $\beta_1 = 0.05$, one extra unit of X gives about 5% more Y, not 5 units.

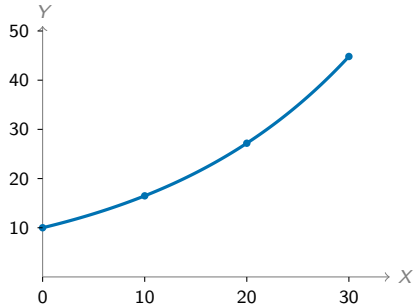
$$\text{Exact change} = 100(e^{0.05} - 1)\% \approx 5.13\%.$$

Why does the curve steepen?

$$Y = e^{\beta_0 + \beta_1 X} \ (u = 0), \qquad \boxed{\frac{dY}{dX} = \beta_1 Y.}$$

For $\beta_1 > 0$, the same percentage increase applies to a growing base: the gain in units gets larger.

Example: $Y = 10e^{0.05X}$



X	0	10	20
Y	10	16.49	27.18
Slope $0.05Y$	0.50	0.82	1.36

The curve is exponential on X, Y axes. It becomes a **straight line** on X, $\log Y$ axes (as in slide 21).

The graph sets $u = 0$. For $\beta_1 < 0$, exponential decay approaches zero and becomes less steep.

Log-log: The Elasticity Determines the Shape

Model and change (holding u fixed):

$$\log Y = \beta_0 + \beta_1 \log X + u,$$

$$\Delta \log Y = \beta_1 \Delta \log X \quad \Rightarrow \quad \frac{\Delta Y}{Y} \approx \beta_1 \frac{\Delta X}{X}.$$

A 1% increase in X:

$$\frac{\Delta X}{X} = 0.01 \quad \Rightarrow \quad 100 \frac{\Delta Y}{Y} \approx \beta_1.$$

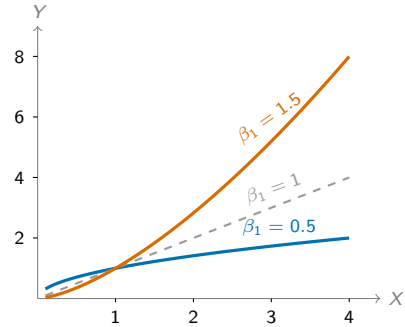
So Y changes by approximately β_1%. Here β_1 is the **elasticity**.

Shape in levels: with $A = e^{\beta_0} > 0$ and $u = 0$,

$$Y = AX^{\beta_1}, \qquad \boxed{\frac{dY}{dX} = A\beta_1 X^{\beta_1 - 1}.}$$

- $0 < \beta_1 < 1$: rises and flattens.
- $\beta_1 = 1$: straight line through the origin.
- $\beta_1 > 1$: rises and steepens.

Examples with $A = 1$: $Y = X^{\beta_1}$

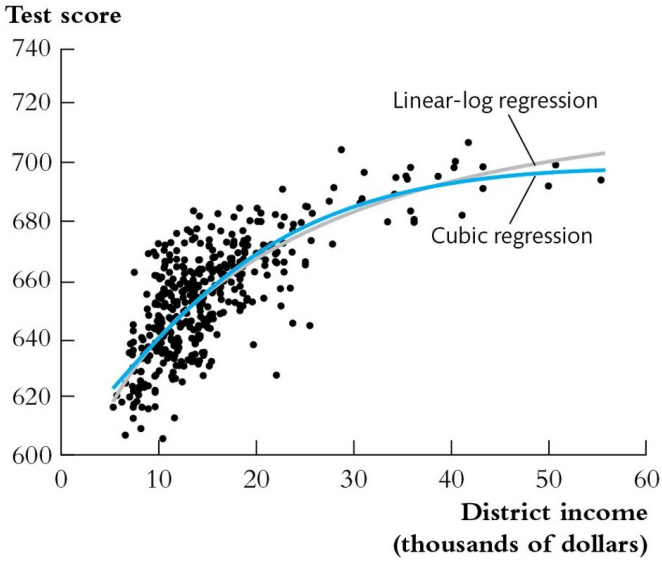


For $\beta_1 = 0.5$, a 1% increase in X gives about 0.5% more Y; for $\beta_1 = 1.5$, about 1.5% more Y.

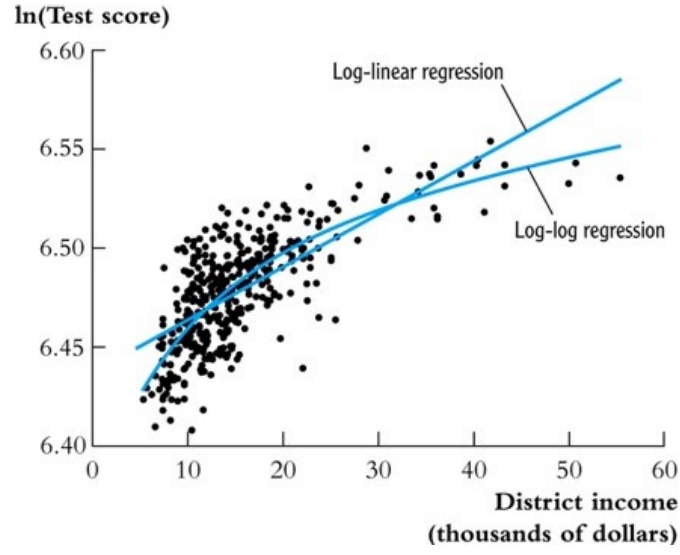
All three become straight lines on $\log X, \log Y$ **axes**, with slopes β_1.

Graphs use the original X, Y axes. If $\beta_1 < 0$, Y decreases and flattens; if $\beta_1 = 0$, it is constant.

(Linear-)Cubic vs. Linear-Log



Log-Linear vs. Log-Log



Interactions and heterogenous effects

- In the California Schools example, at the beginning of the course, we were interested in knowing the impact of the Share of Subsidized Meals on TestScores
- We saw that the Share of Subsidized Meals was, on average, negatively associated with TestScores
- We said that the impact of the Share of Subsidized Meals on on TestScores could be cofounded by local Educational Attainment
- But we might be interested in knowing the *heterogenous* effects of the Share of Subsidized Meals depending on different levels of Educational Attainment
- Is the effect of subsidizing meals still associated with lower test scores, even when local educational attainment is low?

Interactions between two dummies

Consider the following specification

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 (D_{1i} \times D_{2i}) + u_i$$

where D_{1i} and D_{2i} are dummy variables.

- (Comparison category) $D_{1i} = 0$ and $D_{2i} = 0 \Rightarrow \mathbb{E}(Y \mid D_{1i} = 0, D_{2i} = 0) = \beta_0$
- When $D_{1i} = 1$ but $D_{2i} = 0 \Rightarrow \mathbb{E}(Y \mid D_{1i} = 1, D_{2i} = 0) = \beta_0 + \beta_1$
- When $D_{1i} = 0$ but $D_{2i} = 1 \Rightarrow \mathbb{E}(Y \mid D_{1i} = 0, D_{2i} = 1) = \beta_0 + \beta_2$
- Full model when $D_{1i} = 1$ and $D_{2i} = 1 \Rightarrow \mathbb{E}(Y \mid D_{1i} = 1, D_{2i} = 1) = \beta_0 + \beta_1 + \beta_2 + \beta_3$

β_3 can be interpreted as the incremental effect to D_1 (resp. D_2) when $D_2 = 1$ (resp. $D_1 = 1$)

Interactions between two dummies

Dependent variable: TestScore

Regressor	Coefficient
H	-62.37116
L	-42.54073
$H \times L$	24.82402
Intercept	796.65

$n = 500$ $R^2 = 0.3605$ $\text{SER} = 48.368$

$H =$ `high_share_meals` indicates the high subsidized-meal-share group; $L =$ `low_educ` indicates the low educational-attainment group. Each dummy is 1 for its named group and 0 otherwise.

Interactions between a dummy and a continuous variable

How to think about dummies in the presence of a continuous variables? First, consider the simple case without interaction:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 D_{1i} + u_i$$

where D_{1i} is a dummy and X_{1i} is a continuous variable.

The regression line when $D_1 = 0$ reads

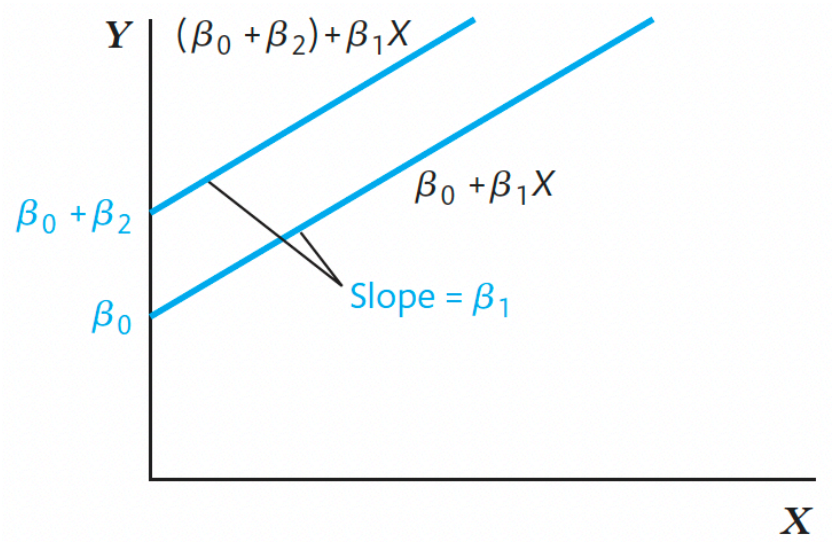
$$\mathbb{E}(Y \mid D_{1i} = 0, X_{1i}) = \beta_0 + \beta_1 X_{2i}$$

While the regression line when $D_1 = 1$ reads

$$\mathbb{E}(Y \mid D_{1i} = 1, X_{1i}) = (\beta_0 + \beta_2) + \beta_1 X_{1i}$$

The intercepts are different but the slopes are the same!

Interactions between a dummy and a continuous variable



Interactions between a dummy and a continuous variable

This framework can be extended to the interaction between a dummy and a continuous variable. Next, consider

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 (D_{1i} \times X_{1i}) + u_i$$

where D_{1i} is a dummy and X_{1i} is a continuous variable.

The regression line when $D_1 = 0$ reads

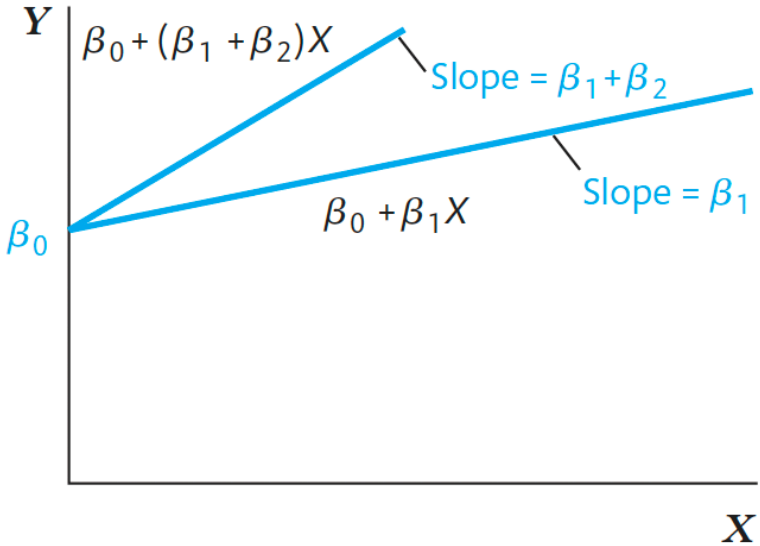
$$\mathbb{E}(Y \mid D_{1i} = 0, X_{1i}) = \beta_0 + \beta_1 X_{1i}$$

While the regression line when $D_1 = 1$ reads

$$\mathbb{E}(Y \mid D_{1i} = 1, X_{1i}) = \beta_0 + \ (\beta_1 + \beta_2) X_{1i}$$

The intercept is the same, but slopes are different!

Interactions between a dummy and a continuous variable



Interactions between a dummy and a continuous variable

Next consider the fully interacted model:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 X_{1i} + \beta_3 (D_{1i} \times X_{1i}) + u_i$$

where D_{1i} is a dummy and X_{1i} is a continuous variable.

The regression line when $D_1 = 0$ reads

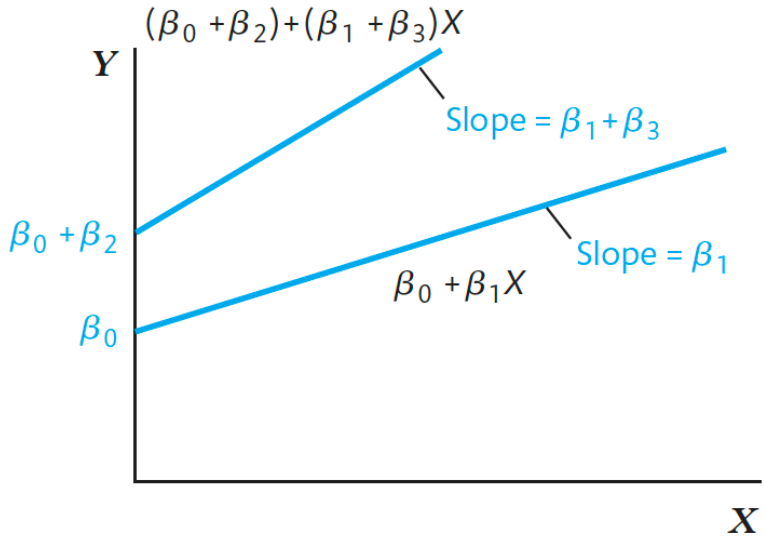
$$\mathbb{E}(Y \mid D_{1i} = 0, X_{1i}) = \beta_0 + \beta_2 X_{1i}$$

While the regression line when $D_1 = 1$ reads

$$\mathbb{E}(Y \mid D_{1i} = 1, X_{1i}) = (\beta_0 + \beta_1) + (\beta_2 + \beta_3) X_{1i}$$

The intercepts and slopes depend on the groups defined by D_1!

Interactions between a dummy and a continuous variable



Interactions between a dummy and a continuous variable

How to discriminate between the last 3 models?

Run the fully interacted model

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 X_{1i} + \beta_3 (D_{1i} \times X_{1i}) + u_i$$

- **Test if the two lines are the same** — (In other words, does the interaction matter?) Compute the F-statistic testing the joint hypothesis that $\beta_1 = \beta_3 = 0$ (Chow Test)
- **Test if the two lines have the same slope** — Compute the t-statistic testing the simple hypothesis that $\beta_3 = 0$
- **Test if the two lines have the same intercept** — Compute the t-statistic testing the simple hypothesis that $\beta_1 = 0$

Interactions between a dummy and a continuous variable

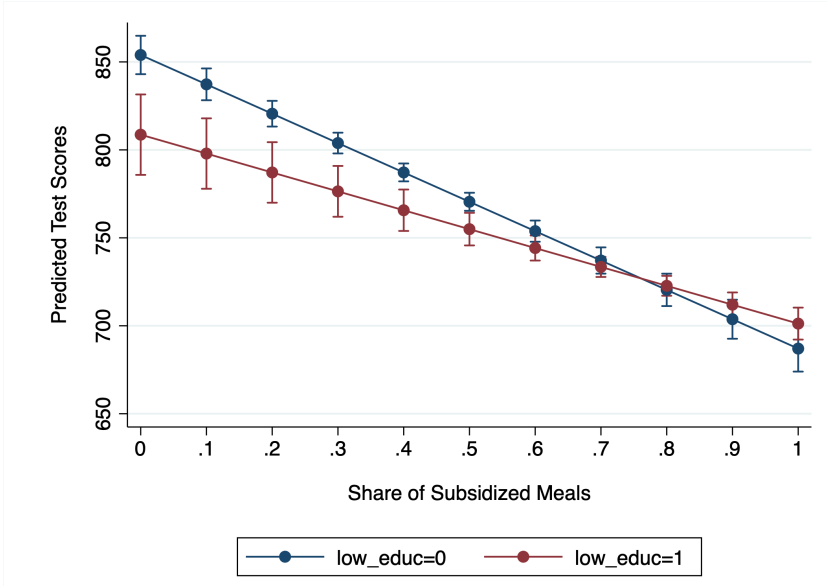
Dependent variable: TestScore

Regressor	Coefficient
S	-162.5841
L	-46.65102
$S \times L$	56.57384
Intercept	853.1778

$n = 500$ $R^2 = 0.5143$ SER $= 42.153$

$S =$ `frpm_frac_s` is the share of students eligible for free or reduced-price meals. $L =$ `low_educ` is 1 for the low educational-attainment group and 0 otherwise.

Interactions between a dummy and a continuous variable



Interactions between two continuous variable

Dependent variable: TestScore

Regressor	Coefficient
S	-245.3094
E	-159.7155
$S \times E$	161.8956
Intercept	944.4173

$n = 500$ $R^2 = 0.5325$ $\text{SER} = 41.358$

$S =$ `frpm_frac_s` is the share of students eligible for free or reduced-price meals. E is the education measure (`ed_frac_sc_lesscollege`).

Wrapping up: Linear vs. Non-Linear Models

In econometrics, we initially favor linear models for their simplicity:

- Linear models assume a constant relationship between variables (e.g., a one-unit increase in X leads to a fixed change in Y).
- However, economic relationships are often complex and dynamic, and this assumption *might not hold in reality*.
- Non-linear models, in contrast, allow for more flexible and *realistic* relationships. They can capture intricate economic behaviors that linear models cannot.

In practice, numerous economic phenomena exhibit non-linear characteristics:

- For example, consider luxury goods. The demand for such goods may not increase linearly with income. As income rises, the demand might accelerate.
- Using linear models in such cases can lead to biased and inaccurate results because they fail to capture these *nuanced relationships*.
- Recognizing non-linear patterns in data is essential for *accurately* modeling and understanding real-world economic dynamics.

The Importance of Interactions

Interactions in econometrics refer to how the effect of one variable depends on the level of another:

- Non-linear interactions can be critical in economic analysis. They enable us to explore *how the relationship between two variables changes as one variable varies*.
- For instance, think about the interaction between interest rates and GDP. The impact of interest rates on investment may vary at different levels of GDP. In times of economic growth, a change in interest rates might have a stronger or weaker effect on investment.
- By incorporating non-linear interactions, we gain a more nuanced understanding of the *interplay between variables*, making our models more realistic and insightful.

$\longrightarrow$ Choosing the right functional form may not only improve your predictive power, it can also have large policy implications!

Material I

- **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 8.